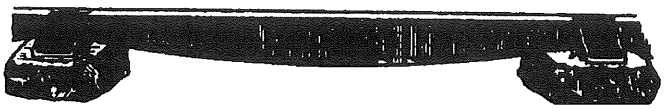
INTRODUCTION TO SIGNALLING

4

TRACKWORK



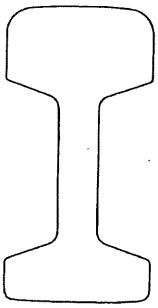
A.J.R.Rowbotham

INTRODUCTION

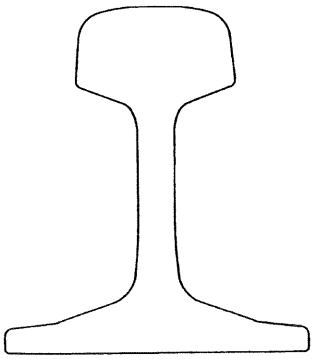
The railway track provides a path on which trains run, and which directs trains in the appropriate direction at junctions. Basically the track consists of steel *rails* on which the trains run, supported on *sleepers* which hold the rails in position. The sleepers are in turn held in position by *ballast*.

RAILS

Rails exist in two designs with different cross sectional shapes. For many years a rail section known as *Bullhead* was used. One of the ideas behind this design was that when it became worn it could be turned over and the other surface used. However this was found to be impractical as the underside of the rail became damaged with time, and was found to have an uneven surface when it was turned. The head of the rail was then made deeper to allow for more wear.



The modern design now almost universally used is known as *Flatbottom.*

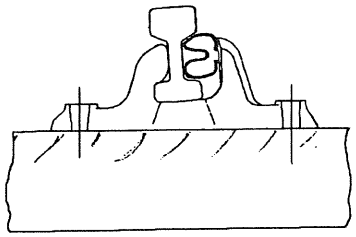


Both designs of rail are available in different weights, the heavier the traffic the more substantial the rail section needed. A very common flat bottom rail in use all over Europe is known as UIC 54. UIC stands for Union Internationale des Chemins de Fer (i.e. International Union of Railways), and the figure '54' is the mass in kg per metre. This section is almost identical to the British Standard BS110A rail, the figure '110' being the weight in lb. per yard.

SLEEPERS

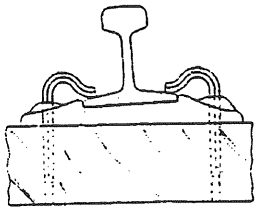
The rails are held in place the correct distance apart by sleepers, which are arranged transversely under the rails. For many years sleepers were cut out of wood, but modern sleepers are of reinforced concrete, which has a substantially longer life. Steel sleepers are also encountered, and have recently been installed in the UK.

Bullhead rail is held in substantial cast *chairs*, which are bolted to the sleepers. Each chair has a wedge shaped wooden or steel key to hold the rail firmly in place.

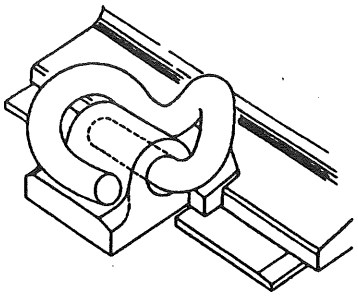


Notice that the rails are not vertical, they slope towards each other slightly. This type of track needs considerable maintenance, it being necessary for a lineman to walk the track every few days, knocking the chocs in firmly. Recently spring steel clips, needing less maintenance, have been used instead of the wooden keys, as is shown in the diagram above.

Flat bottom rail is supported on simple plates, and on the original designs, the rails were held in place by elastic spikes.



An alternative to the spikes, and now very widely used, is the 'Pandrol' clip, a piece of steel rod, bent into an almost indescribable shape, that acts as a spring and firmly holds the rail in place.



Other forms of track support include concrete blocks under each rail held the correct distance apart by metal bars, and paved track. Paved track consists of a continuous concrete slab (laid by a slip form paving machine) on to which the rails are fixed. It has the big advantage over sleepered track in that it needs much less vertical height. It is therefore particularly used in

tunnels where the line is being electrified. To gain height for the overhead line equipment it might well have been necessary to rebuild the tunnel, if it were not for paved track. However, it is very expensive, so its use is restricted.

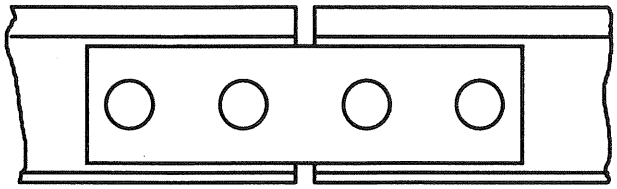
Yet another technique was used in the channel tunnel, which has probably the heaviest traffic of any railway. A flat floor of concrete was laid in the bottom of the tunnel. The rails were then brought in, fitted with concrete blocks at regular intervals. The rails were placed on the concrete base and carefully aligned and levelled. Concrete was then poured in to hold the track in place. If this was literally done, the track system would be a solid mass and crack up in a very short time. This problem was solved by fitting a rubber boot round each concrete block before the rails were brought into the tunnel. These boots provide sufficient resilience when the track is concreted in to prevent the break-up under traffic conditions.

BALLAST

The sleepers are held in place by ballast, which consists of stone chips. This needs to be packed around and under the sleepers to hold them firmly, but resiliently, in place. Today *Tamping Machines*, designed to pack the ballast between and under the sleepers make track maintenance very much less labour intensive. Care must be taken with these massive machines to ensure that any signalling cables in the vicinity of the rails are not damaged.

JOINTS

Rails are manufactured in fairly short lengths and need to be joined together by fishplates to form continuous track. Fishplates consists of metal plates bolted through on each side of the rail.



Fishplates need a lot of maintenance - the bolts must be tight enough to hold the rails together firmly, but not too tight to prevent longitudinal movement due to the expansion and contraction of the rails resulting from temperature change.

In order to reduce maintenance, the Civil Engineer likes to weld the rails into long lengths, forming *jointless* (or *welded*) *track*, and eliminate the fishplates. . Special sliding joints are fitted at the extremities that take up some of the possible movement. When the track is first laid it is actually stretched in such a way that it would not be in stress at a temperature of 27°. Consequently, at normal temperatures the rails will actually be in tension. Only at temperatures above 27° will the rail be in compression. In extremely hot weather this compression may tend to distort the track, and jointless track needs to be laid very carefully with extra ballast to maintain stability. However once laid, it lasts a very long time with minimum maintenance.

For track circuit purposes *insulated joints* are needed by the Signal Engineer. In jointed track, insulation is incorporated in the fishplate items, and an insulating plate of the appropriate cross-sectional area is fitted between the rail ends to prevent any contact in hot weather. For insulated joints in 'jointless track' the most reliable way of making them is to take two short lengths of rail and make a *glued* (and insulated) *joint* between them under carefully controlled factory conditions. The joint is then taken out and welded into the track where required.

GAUGE

The distance between the railheads is known as the track *gauge*. In England, Scotland and Wales the gauge is 4 ft 8½" and is known as *Standard Gauge*. It is claimed that this strange measurement is a result of the distance between the wheels of Roman chariots? In metric terms this gauge is 1,435 mm. Luckily the same gauge is found in the closest countries in mainland Europe, so there is no problem in trains running through the Channel Tunnel. However, the railways on the Iberian Peninsular are to the wider gauge of 1668 mm and this of course produces severe problems for through trains running between France and Spain. For many years every wagon and carriage had to be jacked up and have its bogies or wheels changed. Nowadays on more modern stock the axles are designed so the wheels can slide, under carefully controlled conditions, on the axles while traversing a section of track where the gauge changes gradually from one to the other. There is now one new standard gauge line in Spain, and plans are being formulated to gradually convert the main lines to standard gauge to improve communication with the rest of Europe. Elsewhere in Europe there are other gauge differences. In the Russian republics and Finland the gauge is 5 ft whilst in Eire/ Northern Ireland it is 5 ft 3".

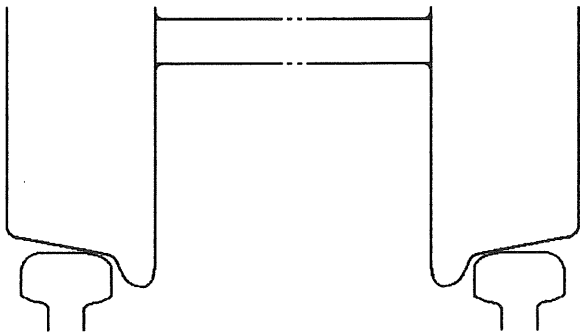
The big problems however occur in countries such as Australia and India where there is a considerable mixture of gauges. In Australia at the time when the railways were built there was no overall railway body, so each state acted independently with New South Wales having standard gauge, Western Australia 3 ft 6" gauge (although the extremely long line to the other States is standard gauge) and South Australia/Victoria at 5 ft 3". Slowly a standard gauge line is being built to join the states together to allow through running. This involves sections of *dual gauge* track, a system with three rails. On one side of the track is one rail, the common rail, while on the other side there are two rails, each at the appropriate distance from the common rail for the two gauges required.

India is another problem country with a mixture of broad Gauge and narrow gauge lines, not separated by their location in the country. Again a huge exercise is being carried out, converting a number of narrow gauge lines into broad gauge lines (which is a difficult and very expensive exercise as more space is needed).

A very common gauge in mountainous areas such as Switzerland is metre gauge, and there is a vast network of strategically important main line metre gauge railways in the alpine area. In winter these lines form virtually the only means of travelling. The Imperial equivalent of metre gauge is the 3 ft 6" gauge mentioned earlier. All the railways in southern Africa are built to this gauge, where it is known as the 'Cape Gauge'.

RAIL/WHEEL INTERFACE

Most people believe that it is the flanges on the wheels that keep the wheels on the rails. This is only true on very sharp curves, when the flanges actually rub along the rails (usually audibly). If the flanges were in contact with the rail continuously then the wear would be intolerable and impractical. The drawing below, which is not to scale, shows the interface between the rail and the wheel tread.



Note that the wheel tread is not parallel with the top of the rail head (although the slope is deliberately exaggerated in the drawing). Considering the case where the axle is tending to move to the left - the left hand wheel will become a larger diameter at the point of contact, whilst the right hand wheel will become a smaller diameter at the point of contact. As both wheels are attached to the same axle, they must be making the same number of revolutions per minute. This difference in effective diameter will cause the axle to move back to the situation where both wheels have the same effective diameter. A lot of research into the wheel/rail interface has resulted in the optimum profile shapes, resulting in very straight running with a minimum of hunting backwards and forwards and significantly reducing rail and flange wear.

SYMBOLS AND NOMENCLATURE

On *Track Plans* (also called *Scheme Plans* or *Numbering Plans*) it is not necessary to show the details of both rails, so a single line depicts a section of track. A *Single Track* is thus represented by one line on the drawing, whilst a *Double Track* is represented by two lines, one for each direction. In the UK the railways 'drive on the left' to match the roads. However. on mainland Europe, inconsistently, some railways drive on the left, whilst others drive on the right. France and Switzerland are among those that drive on the left, whilst Germany (and in fact the Alsace area of France, which was part of Germany when the railways were built), Austria and Spain are amongst those that drive on the right.

Within the area controlled by every signal box, or signal centre, every line must have a unique name, so it can be referred to in instructions or any other paperwork. In a very small box just controlling a stretch of double track, it is sufficient to just designate the lines as 'Up' and 'Down'. Traditionally the up line is for traffic moving towards London, whilst the Down Line is for traffic moving away from London. This definition, of course, becomes somewhat arbitrary if the lines are nowhere near London. On a slightly bigger area, Up and Down Main and Up and Down Branch would be satisfactory. On a big resignalling scheme, where the areas controlled by perhaps 20 old small boxes are to be combined, new names have to be thought up to avoid duplication, usually with some local connection..

The space between the rails is known as *the four foot*, e.g. signalling equipment mounted on the sleepers between the rails is said to be 'in the four foot'. In a similar way, the space between the lines on a double track section is known as *the six foot*. On a quadruple section of track, the space between the two pairs of lines is known as *the ten foot*. These are not the exact measurements, but they are convenient ways of describing the areas. The *Cess* is the area directly adjacent to the outside of the running lines.